Circuit model and solder practice

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Test a circuit representation before making permanent connections.

SUCCESS CRITERIA - I can...

1. Build a complete battery-switch-motor circuit model.
2. Explain an open versus closed path.
3. Document a supervised solder-practice check.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A car prototype is ready for a planned 5 m powered run. Name a safety check before power and a measurement needed to evaluate the run.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $I = \Delta q/\Delta t$; $I = 0$ in an ideal open circuit.
Use the assigned simulation and teacher solder procedure. Disconnect power; hot irons/joints stay in the designated work area.

Worked reasoning: In a simple car circuit, battery positive connects through the switch and motor back to battery negative. A wire directly across the battery bypasses the motor and can create a hazardous short circuit.

Your turn: Draw the intended closed loop and circle a place where an open switch safely breaks it.



Marzano check - circle: ① ② ③

03 Read and interpret the model

Draw your simulated closed circuit and an open-switch version. Predict which runs the motor and why.



Circuit symbols: long/short lines = battery; rectangles = resistors; M = motor.

Marzano check - circle: ① ② ③

04 Apply the physics

In a circuit simulation, a 3 V supply delivers 0.20 A to a motor. Label voltage and current and calculate input power. Make one approved change in the simulation and record what you actually observe.



Marzano check - circle: ① ② ③

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05 Explain your reasoning

Make a two-column record: practice-joint observation and next improvement. Include teacher feedback on the joint and tool handling.

Observed joint / handling	Teacher feedback	Next time
___	___	___
___	___	___

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Parallel branches draw 0.4 A and 0.6 A. Find the current supplied to their junction.
Draw the stated circuit and label quantities relevant to the question.

Draw the complete circuit and label shared nodes.

Marzano check - circle: 1 2 3

07 Exit check

Before soldering a circuit connection, should power be connected or disconnected? State one reason.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

Draw two circuit faults on the blank models: (A) one disconnected lead; (B) a wire that bypasses the motor and connects across the battery. Explain the expected behavior and identify a teacher-approved first check with power disconnected. These are drawings, not faults to build.

A: open lead

B: bypass

Draw each circuit fault; label the motor and battery.

Marzano check - circle: 1 2 3